Simon Oddershede Gregersen

Email	gregersen@cispa.de
Website	https://simongregersen.com
ORCID	0000-0001-6045-5232
DBLP	https://dblp.org/pid/263/1059

Employment

2026 –	**CISPA Helmholtz Center for Information Security**. Tenure-track faculty.
2024 – 2025	**New York University**. Postdoctoral fellow. Advised by Joseph Tassarotti.
2023 – 2024	**Aarhus University**. Postdoctoral researcher. Advised by Lars Birkedal.
2017	**Aarhus University**. Center for Advanced Software Analysis. Student programmer.
2015 – 2017	**CCI Europe A/S (now Stibo DX)**. Student software developer.

Education

2017 – 2023	**PhD in Computer Science**. Aarhus University. Advised by Amin Timay and Lars Birkedal. Thesis: Higher-Order Separation Logic for Distributed Systems and Security.
2017 – 2020	**MSc in Computer Science**. Aarhus University.
2014 – 2017	**BSc in Computer Science**. Aarhus University.

Publications

[13] Philipp G. Haselwarter, Alejandro Aguirre, Simon Oddershede Gregersen, Kwing Hei Li, Joseph Tassarotti, and Lars Birkedal. "Modular Verification of Differential Privacy in Probabilistic Higher-Order Separation Logic". In: *Proc. ACM Program. Lang.* 10.PLDI (2026). DOI: 10.1145/3808311.

[12] Simon Oddershede Gregersen, Chaitanya Agarwal, and Joseph Tassarotti. "Logical Relations for Formally Verified Authenticated Data Structures". In: *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, CCS 2025, Taipei, Taiwan, October 13-17, 2025.* 2025, pp. 1394–1408. DOI: 10.1145/3719027.3744801.

[11] Kwing Hei Li, Alejandro Aguirre, Simon Oddershede Gregersen, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. "Modular Reasoning about Error Bounds for Concurrent Probabilistic Programs". In: *Proc. ACM Program. Lang.* 9.ICFP (2025). DOI: 10.1145/3747514.

[10] Philipp G. Haselwarter, Kwing Hei Li, Alejandro Aguirre, Simon Oddershede Gregersen, Joseph Tassarotti, and Lars Birkedal. "Approximate Relational Reasoning for Higher-Order Probabilistic Programs". In: *Proc. ACM Program. Lang.* 9.POPL (2025). DOI: 10.1145/3704877.

[9] Philipp G. Haselwarter, Kwing Hei Li, Markus de Medeiros, Simon Oddershede Gregersen, Alejandro Aguirre, Joseph Tassarotti, and Lars Birkedal. "Tachis: Higher-Order Separation Logic with Credits for Expected Costs". In: *Proc. ACM Program. Lang.* 8.OOPSLA2 (2024). DOI: 10.1145/3689753.

[8] Simon Oddershede Gregersen, Alejandro Aguirre, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. “Almost-Sure Termination by Guarded Refinement”. In: *Proc. ACM Program. Lang.* 8.ICFP (2024). DOI: 10.1145/3674632.

[7] Alejandro Aguirre, Philipp G. Haselwarter, Markus de Medeiros, Kwing Hei Li, Simon Oddershede Gregersen, Joseph Tassarotti, and Lars Birkedal. “Error Credits: Resourceful Reasoning about Error Bounds for Higher-Order Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 8.ICFP (2024). DOI: 10.1145/3674635.

[6] Amin Timany, Simon Oddershede Gregersen, Léo Stefanesco, Jonas Kastberg Hinrichsen, Léon Gondelman, Abel Nieto, and Lars Birkedal. “Trillium: Higher-Order Concurrent and Distributed Separation Logic for Intensional Refinement”. In: *Proc. ACM Program. Lang.* 8.POPL (2024). DOI: 10.1145/3632851.

[5] Simon Oddershede Gregersen, Alejandro Aguirre, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. “Asynchronous Probabilistic Couplings in Higher-Order Separation Logic”. In: *Proc. ACM Program. Lang.* 8.POPL (2024). DOI: 10.1145/3632868.

[4] Simon Oddershede Gregersen, Johan Bay, Amin Timany, and Lars Birkedal. “Mechanized Logical Relations for Termination-Insensitive Noninterference”. In: *Proc. ACM Program. Lang.* 5.POPL (2021). DOI: 10.1145/3434291.

[3] Léon Gondelman, Simon Oddershede Gregersen, Abel Nieto, Amin Timany, and Lars Birkedal. “Distributed Causal Memory: Modular Specification and Verification in Higher-Order Distributed Separation Logic”. In: *Proc. ACM Program. Lang.* 5.POPL (2021). DOI: 10.1145/3434323.

[2] Morten Krogh-Jespersen, Amin Timany, Marit Edna Ohlenbusch, Simon Oddershede Gregersen, and Lars Birkedal. “Aneris: A Mechanized Logic for Modular Reasoning about Distributed Systems”. In: *Programming Languages and Systems - 29th European Symposium on Programming, ESOP 2020, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2020, Dublin, Ireland, April 25-30, 2020, Proceedings.* 2020. DOI: 10.1007/978-3-030-44914-8_13.

[1] Simon Oddershede Gregersen, Søren Eller Thomsen, and Aslan Askarov. “A Dependently Typed Library for Static Information-Flow Control in Idris”. In: *Principles of Security and Trust - 8th International Conference, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2019, Prague, Czech Republic, April 6-11, 2019, Proceedings.* 2019. DOI: 10.1007/978-3-030-17138-4_3.

Awards

- Distinguished Paper Award at ICFP 2024.
- Carlsberg Foundation Internationalization Fellowship (CF23-0791). DKK 1,020,000 ($\approx$ USD 150,000).

Advising

PhD students

- Aleksey Veresov (2026 –).

Interns

- Yanees Dobberstein (2026 May – July).

MSc Thesis Committees

- Peter Gastaur (2026). Saarland University.
  Thesis: Compiling Distributed Algorithms in Pseudocode into Extended Threshold Automata.

Academic Service

Organizing

2027 Dagstuhl Seminar (no. 202606056). Foundations for Security of Cryptographic Protocols.

2026 Artifact Evaluation Co-Chair. Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA) .

Program Committees

2027 Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA).
Certified Programs and Proofs (CPP).

2026 SPLASH Onward!
Recent Advances in Concurrency and Logic (RADICAL).
Foundations of Computer Security (FCS).
Programming Language Design and Implementation (PLDI).

2025 Recent Advances in Concurrency and Logic (RADICAL).

External Reviewing

Journals JFP (2025), TCS (2024), TOSEM (2023)

Conferences LICS 2025, LICS 2024, CCS 2022, POPL 2022, OOPSLA 2022, ICFP 2022, CSF 2022, CSF 2021, ESOP 2020

Selected Talks

2026 Feb *Building Extensible Program Logics through Effect Handlers.*
MPI-SWS Foundations of Programming group seminar.

2025 Oct *Logical Relations for Formally Verified Authenticated Data Structures.*
New England Systems Verification Day.

2025 Jun *Logical Relations for Formally Verified Authenticated Data Structures.*
The Fifth Iris Workshop.

2025 May *Logical Relations for Formally Verified Authenticated Data Structures.*
New Jersey Programming Languages and Systems Seminar.

2025 Apr *Logical Relations for Formally Verified Authenticated Data Structures.*
VU Amsterdam PLSec reading group.

2024 Apr *Trillium: Intensional Refinement in Higher-Order Separation Logic.*
New England Systems Verification Day.

2023 Jul *Asynchronous Probabilistic Couplings in Higher-Order Separation Logic.*
Bristol Programming Languages Research group seminar.

2023 Jul *Asynchronous Probabilistic Couplings in Higher-Order Separation Logic.*
Workshop on Verification of Probabilistic Programs (VeriProP).

2022 May *Trillium: History-Sensitive Refinement in Separation Logic.*
The Second Iris Workshop.

2020 Nov *Mechanized Logical Relations for Termination-Insensitive Noninterference.*
Chalmers ProgLog/Security seminar.